

MACHINE LEARNING JOURNEY — DEEP DIVE NOTES

Deep Learning, Neural Networks & Applied AI

Beginner-to-Intermediate Notes with Real Diagrams, Illustrations & Interview Prep

Topics Covered:

1. Deep Learning
2. ML Training Process
3. Neural Network Architecture (Input / Hidden / Output Layers)
4. Shallow vs Deep Learning
5. Applied AI vs Generalized AI
6. Narrow Tasks vs Multiple Coordinated Tasks
7. Why Do We Need AI Today?
8. Data in AI
9. Hardware in AI
10. Cloud-Based AI

Prepared for Coursera Machine Learning Course — Session 2 Notes

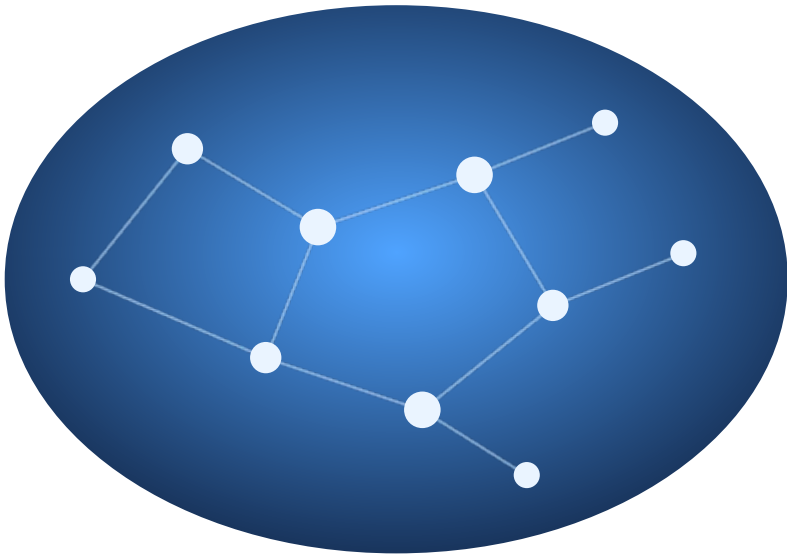
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Deep Learning

1. Introduction — Definition & Overview

Deep Learning is a specialised branch of Machine Learning that uses **artificial neural networks with many stacked layers** ("deep" = many layers) to automatically learn complex patterns directly from large amounts of raw data — such as images, audio, or text — without a human manually defining what features to look for.



Deep Learning = many connected layers of artificial "neurons"

Illustration: a deep neural network is loosely inspired by how neurons connect in the human brain.

2. In-Depth Explanation — How Deep Learning Works

A deep learning system is built from layers of small mathematical units called **neurons**. Each neuron receives numbers, multiplies them by learned "weights," adds them up, and passes the result through the next layer. Early layers learn very simple patterns (like edges in an image); deeper layers combine these into more complex ideas (shapes, then objects, then full scenes).

Layer Depth	What It Tends to Learn (Image Example)
Layer 1 (near input)	Simple edges, colours, contrast changes
Middle layers	Shapes, textures, small parts (eyes, wheels, corners)
Deep layers (near output)	Whole objects (a face, a car, a cat)

3. How It Works — Step by Step

1. Raw data (e.g., an image) is fed into the **input layer** as numbers (pixel values).
2. Data passes through many **hidden layers**, each transforming it slightly using weights learned during training.
3. The **output layer** produces a final result (e.g., "cat: 92% confidence").
4. The network compares its answer to the correct label and calculates an **error**.
5. Using a process called **backpropagation**, the network adjusts its internal weights slightly to reduce that error.
6. This repeats across thousands or millions of examples until the network becomes accurate.

4. Real-World Examples

Image Recognition — Identifying objects, faces, or medical conditions in photos and scans.

Speech Recognition — Converting spoken words into text (voice assistants, subtitles).

Language Translation — Translating text between languages while preserving meaning.

Self-Driving Perception — Detecting lanes, pedestrians, and obstacles from camera feeds in real time.

Generative AI — Creating new text, images, or audio (chatbots, art generators, music tools).

5. Key Points

- Deep Learning = Machine Learning + many stacked layers of artificial neurons.
- It learns features automatically — no need to manually tell it what to look for.
- Needs large datasets and strong computing power (GPUs) to train well.
- Powers most modern breakthroughs: image recognition, voice assistants, chatbots, self-driving cars.

6. Advantages & Limitations

Advantages	Limitations
Learns complex patterns without manual feature design	Requires huge amounts of labelled data
State-of-the-art accuracy on images, speech, text	Needs powerful, often expensive hardware
Improves further as more data is added	Hard to explain exactly why it made a decision ("black box")
Reusable across many domains (vision, audio, language)	Long training times for very large models

7. Interview Questions

Q1. How is Deep Learning different from traditional Machine Learning?

A: Traditional ML often needs a human to hand-pick "features" from the data; Deep Learning automatically learns useful features on its own through multiple layers of neurons.

Q2. Why does Deep Learning need so much data?

A: Because it has millions of internal weights to tune correctly; with too little data, it cannot learn reliable patterns and may just memorise examples instead of generalising.

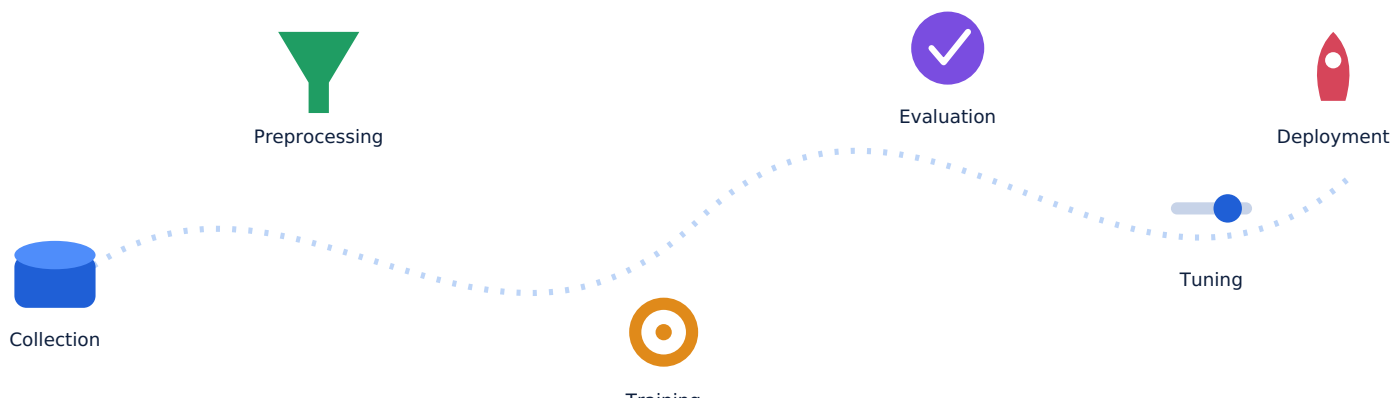
Q3. What is backpropagation, in simple terms?

A: A process where the network checks how wrong its prediction was, then works backward through its layers, slightly adjusting weights to reduce that error next time.

The Machine Learning Training Process

1. Introduction

Training is the journey a model goes through to turn raw data into a working, usable predictor. It has six main stages: **Data Collection** → **Data Preprocessing** → **Model Training** → **Model Evaluation** → **Hyperparameter Tuning** → **Deployment**.



The ML training journey: each stage feeds into the next, but real projects often loop back for improvement.

2. In-Depth Explanation of Each Stage

Data Collection

Gathering raw examples relevant to the problem — from databases, sensors, user activity logs, images, or public datasets. The quality and variety of this data sets the ceiling for how good the final model can be.

Data Preprocessing

Cleaning and organising the raw data: removing duplicates, handling missing values, converting text/images into numbers, and splitting the dataset into **training** and **testing** portions.

Model Training

The chosen algorithm studies the training data repeatedly, adjusting its internal parameters (weights) to reduce prediction errors.

Model Evaluation

The trained model is tested on data it has never seen, using metrics like accuracy, precision, or error rate, to check if it generalises well rather than just memorising.

Hyperparameter Tuning

Settings that control how the model learns (e.g., learning speed, number of layers) are adjusted and re-tested to squeeze out better performance.

Deployment

The finished, tested model is integrated into a real application (a website, app, or service) so it can make live predictions for real users.

3. Real-World Examples

Spam Filter — Collects millions of labelled emails → cleans text → trains a classifier → evaluates accuracy → tunes thresholds → deploys inside Gmail.

Product Recommendation — Collects purchase history → preprocesses into user-item patterns → trains a recommender model → evaluates click-through rate → deploys on the shopping site.

Credit Scoring — Collects financial history → cleans/normalises data → trains a risk model → evaluates on past loan outcomes → deploys into the bank's approval system.

4. Key Points

- Training is a cycle, not a straight line — teams often loop back to earlier stages to improve results.
- Garbage in, garbage out: poor data collection/cleaning limits the entire pipeline.
- Evaluation must always use data the model has not already seen.

5. Interview Questions

Q1. Why is data preprocessing considered so important?

A: Because raw, messy, or incomplete data can mislead the model into learning wrong patterns; clean data is essential for reliable training.

Q2. What is the difference between model training and model evaluation?

A: Training is when the model learns patterns from known data; evaluation is testing the model's performance on separate, unseen data to check real-world reliability.

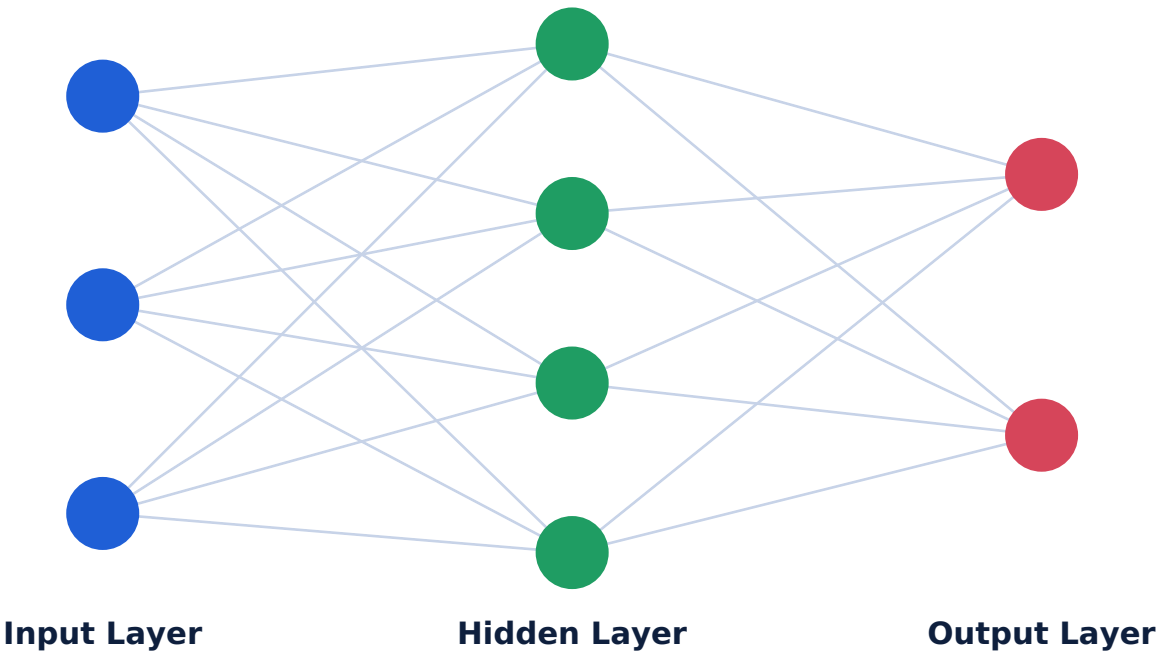
Q3. What is hyperparameter tuning?

A: The process of adjusting the model's external settings (like learning rate or number of layers) to improve performance, as opposed to the weights the model learns automatically.

Neural Network Architecture — Input, Hidden & Output Layers

1. Introduction

A neural network is organised into layers of neurons: an **Input Layer** (receives raw data), one or more **Hidden Layers** (do the actual pattern-finding), and an **Output Layer** (gives the final answer). Neurons in one layer connect to neurons in the next layer through weighted connections.



A basic neural network: every neuron in one layer connects to every neuron in the next via a "weight."

2. In-Depth Explanation

Layer	Role	Example (Predicting House Price)
Input Layer	Receives raw feature values, one neuron per feature	Size, location, number of rooms (3 input neurons)
Hidden Layer(s)	Combines inputs in complex ways to detect patterns	Learns combinations like "big house in good area"
Output Layer	Produces the final prediction	Predicted price (1 output neuron)

Key Terms: Weights, Neurons, Activations

- **Neuron** — a small unit that takes inputs, combines them, and produces one output number.
- **Weight** — a number attached to each connection that controls how much influence one neuron has on the next; these are what the network "learns."
- **Activation** — a function applied inside each neuron that decides how strongly it "fires," allowing the network to learn non-simple, curved patterns instead of only straight-line relationships.
- **Bias** — an extra adjustable number added to a neuron's calculation, giving the network more flexibility to fit patterns accurately.

3. How It Works — Step by Step

1. Each input neuron holds one raw feature value (e.g., house size = 1500).
2. Every connection multiplies the input value by its weight.
3. Each hidden neuron adds up all the weighted values it receives, plus a bias.
4. An activation function transforms this sum into the neuron's output signal.
5. This output flows forward to the next layer, repeating the process.
6. The output layer produces the final prediction.

4. Real-World Examples

Handwriting Recognition — Input neurons = pixel values of a digit image; output neurons = probability of each digit 0-9.

Email Spam Detection — Input neurons = word frequency counts; output neuron = probability the email is spam.

5. Key Points

- Data flows forward: Input → Hidden → Output.
- Weights are the "knowledge" the network learns during training.
- More hidden neurons/layers allow more complex patterns to be captured — but also require more data and computing power.

6. Interview Questions

Q1. What is the role of the hidden layer in a neural network?

A: It transforms the raw input into increasingly abstract, useful representations that help the output layer make an accurate final prediction.

Q2. What would happen without an activation function?

A: The entire network would behave like one simple straight-line (linear) equation, no matter how many layers it has, and could not learn complex, curved real-world patterns.

Q3. What exactly does a neural network "learn" during training?

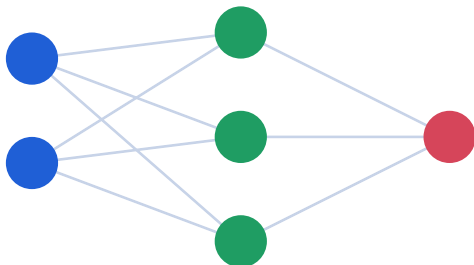
A: It learns the numeric values of its weights and biases — the settings that determine how strongly each neuron influences the next layer.

Shallow vs Deep Learning

1. Introduction

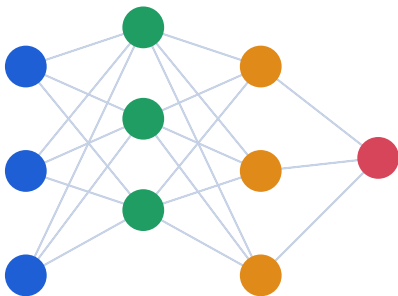
A **shallow network** has just one (or zero) hidden layers — simple and fast, good for straightforward problems. A **deep network** stacks many hidden layers, allowing it to learn far more complex, abstract patterns — at the cost of needing more data and computing power.

Shallow Network



1 hidden layer — good for simple, well-defined problems

Deep Network



Many hidden layers — captures complex, abstract patterns

Shallow networks have 1 hidden layer; deep networks stack several, each learning increasingly abstract features.

2. Comparison Table

Aspect	Shallow Network	Deep Network
Hidden Layers	0-1	Many (often 10s-100s)
Pattern Complexity	Simple, mostly straightforward relationships	Complex, abstract, hierarchical patterns
Data Needed	Smaller datasets often enough	Requires large datasets
Training Speed	Fast	Slower, more computationally expensive
Best Use Case	Simple prediction tasks (e.g., basic price estimates)	Images, speech, language, complex vision tasks
Hardware Needs	A normal CPU is often enough	Usually needs GPUs/TPUs

3. Real-World Examples

Shallow Network Example — Predicting a student's exam score from study hours and attendance (a simple, clear relationship).

Deep Network Example — Recognising a cat in a photo, where the pattern (fur, ears, shape, pose, lighting) is far too complex for a shallow model to capture well.

4. Key Points

- "Depth" refers to the number of hidden layers stacked between input and output.
- More depth = more capacity to learn complex patterns, but also more risk of overfitting without enough data.
- Not every problem needs a deep network — simple problems are often solved just as well (and faster) by shallow ones.

5. Interview Questions

Q1. When would you choose a shallow network over a deep one?

A: When the problem is simple, the dataset is small, and fast training/inference matters more than squeezing out maximum accuracy.

Q2. Why can deep networks overfit more easily?

A: Because they have far more parameters (weights) to learn, so with too little data they can start memorising training examples instead of learning general patterns.

Applied AI vs Generalized AI

1. Introduction

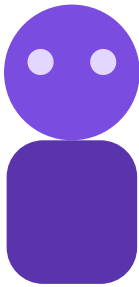
Applied AI (also called Narrow AI) is designed to do one job extremely well. **Generalized AI** (General AI / AGI) is a theoretical form of AI that could understand, learn, and perform *any* intellectual task a human can — it does not exist yet.

Applied AI (Narrow)



One job: e.g. voice assistant

Generalized AI (Theoretical)



Any task a human can do — not yet real

Applied AI = specialist. Generalized AI = an all-purpose thinker, still theoretical today.

2. Key Differences

Aspect	Applied AI (Narrow)	Generalized AI (AGI)
Scope	One specific task	Any intellectual task
Exists Today?	Yes — this is all current AI	No — purely theoretical/research goal
Learning Transfer	Cannot apply knowledge outside its trained task	Could apply knowledge across completely different tasks
Examples	Siri, spam filters, chess engines, face unlock	None yet — a long-term research aspiration

3. Real-World Examples

- Applied AI** — A recommendation engine is excellent at suggesting movies but cannot drive a car or diagnose an illness.
- Applied AI** — A self-driving car's perception system is excellent at reading roads but cannot write an essay.
- Generalized AI (hypothetical)** — A single system that could learn to cook, drive, write code, and hold a genuine conversation — all with the same underlying "mind," the way a human can.

4. Key Points

- Every AI product you use today — no matter how impressive — is Applied/Narrow AI.
- Generalized AI remains a research goal, with open questions about whether/when it will be achieved.
- Don't confuse "sounds smart in conversation" with "generally intelligent" — even very fluent AI systems are still narrow underneath.

5. Interview Questions

- Q1. Is a chatbot that can discuss many topics an example of Generalized AI?**

A: No — even a broad-sounding chatbot is still Applied/Narrow AI. It's trained on huge amounts of text to sound versatile, but it can't truly transfer learning to unrelated real-world tasks like a human can.
- Q2. Why hasn't Generalized AI been achieved yet?**

A: True general intelligence requires reasoning, transferring knowledge across unrelated domains, and common-sense understanding — capabilities researchers have not been able to fully replicate yet.

Narrow Tasks vs Multiple Coordinated Tasks

1. Introduction

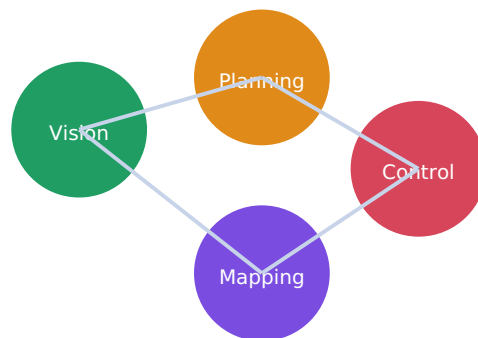
A **narrow-task AI** handles a single, well-defined job in isolation. A system built for **multiple coordinated tasks** combines several narrow-AI components working together to accomplish something bigger — each part still narrow on its own, but coordinated as a team.

Narrow Task



One isolated skill

Multiple Coordinated Tasks



Several narrow skills working as a team

A self-driving car doesn't have one "brain" — it coordinates several narrow AI systems together.

2. Explanation

Even highly capable AI systems, like self-driving cars or robots, are usually not one single intelligence. They are a **coordinated collection of narrow AI models** — one for seeing the road, one for predicting other cars' movement, one for planning a route, one for controlling the steering — all communicating together to produce one smooth behaviour.

3. Real-World Examples

Narrow Task Example — A spam filter does exactly one job: classify email as spam or not spam.

Multiple Coordinated Tasks Example — A self-driving car coordinates vision (seeing), prediction (guessing what pedestrians will do), planning (choosing a path), and control (steering/braking) simultaneously.

Multiple Coordinated Tasks Example — A warehouse robot coordinates navigation, object recognition, and grip-strength control to pick and move packages safely.

4. Key Points

- Coordinating multiple narrow models is NOT the same as Generalized AI — each part is still narrow and task-specific.
- This "team of specialists" approach is how most advanced real-world AI systems are actually built today.

5. Interview Questions

Q1. Does a self-driving car count as Generalized AI since it does many things at once?

A: No — it is a coordinated system of multiple narrow AI models (vision, planning, control), each trained for its own specific task, not one general intelligence.

Why Do We Need AI Today?

1. Introduction

AI has become essential because the modern world produces far more data and far more repetitive, time-critical decisions than humans alone can process — AI helps automate work, uncover insights, and make faster, more consistent decisions at massive scale.



Automation

Handles repetitive tasks non-stop, without fatigue.



Handling Big Data

Processes millions of records instantly.



Better Decisions

Finds patterns humans might miss, faster.



Improving Accuracy

Reduces human error in critical tasks.

2. Explanation

Three forces have made AI necessary rather than optional: the explosion of digital data, the need for round-the-clock automated systems, and the demand for personalised, real-time decisions (like fraud alerts or route changes) that are too fast for manual review.

3. Real-World Examples

Automation — Customer support chatbots handle thousands of queries simultaneously, 24/7.

Big Data Handling — Social media platforms analyse billions of posts daily to moderate content and rank feeds.

Real-Time Decisions — Banks approve or flag transactions within milliseconds using AI risk models.

4. Key Points

- AI is needed where tasks are too large, too fast, or too repetitive for humans alone.
- It complements human judgement rather than fully replacing it in most real applications.

5. Interview Questions

Q1. Give one reason AI adoption has accelerated in the last decade.

A: A massive increase in available digital data combined with cheaper, more powerful computing hardware made training accurate AI models practical at scale.

Data in AI

1. Introduction

Data is the fuel of every AI system. It generally falls into two broad categories: **Structured Data** (neatly organised, like spreadsheets) and **Unstructured Data** (raw and free-form, like images, audio, or free text).



Structured Data
Tables, spreadsheets, databases (rows & columns).



Text Data
Emails, articles, chat messages, reviews.



Image Data
Photos, scans, video frames.



Audio Data
Voice recordings, music, sound signals.

2. Structured vs Unstructured Data

Aspect	Structured Data	Unstructured Data
Format	Organised rows & columns	Free-form (text, images, audio, video)
Example	A sales spreadsheet	Customer support voice recordings
Ease of Analysis	Easier for traditional tools	Needs AI/ML (especially Deep Learning) to interpret
Storage	Relational databases, spreadsheets	Data lakes, cloud object storage

3. Real-World Examples

- Structured** — A bank's transaction table: date, amount, account number, category.
- Unstructured (Text)** — Millions of product reviews used to train a sentiment-analysis model.
- Unstructured (Image)** — X-ray scans used to train a medical diagnosis model.
- Unstructured (Audio)** — Call-centre recordings used to train a speech-to-text system.

4. Key Points

- Most of the world's data today is unstructured — this is exactly why Deep Learning became so important.
- Good data quality and labelling matter more than raw data volume alone.

5. Interview Questions

Q1. Why is unstructured data harder to work with than structured data?

A: It has no fixed format or predictable layout, so extracting meaningful features from it (e.g., from an image or an audio clip) requires more sophisticated models, typically deep neural networks.

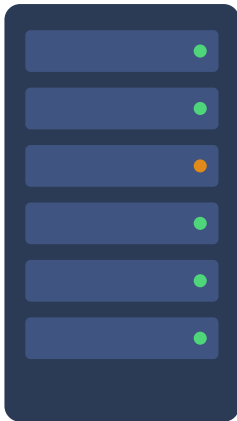
Hardware in AI

1. Introduction

Training modern AI models involves billions of repeated mathematical calculations. Specialised hardware — **GPUs** (Graphics Processing Units) and **TPUs** (Tensor Processing Units) — perform these calculations in parallel, making them far faster than a normal computer's CPU for AI workloads.



GPU



Server Rack



TPU / Accelerator

GPUs and TPUs run thousands of small calculations in parallel — ideal for training neural networks quickly.

2. Role of GPUs and TPUs

Hardware	Best For	Why It Matters for AI
CPU	General-purpose computing	Good at sequential tasks, but slow for massive parallel math
GPU	Parallel matrix/vector math	Thousands of small cores can train neural networks far faster than a CPU
TPU	Deep learning specifically	Custom-built chips (by Google) optimised purely for neural network operations

3. Why Powerful Hardware Is Needed

- Training a deep neural network involves adjusting millions (sometimes billions) of weights repeatedly.
- GPUs/TPUs perform many of these calculations simultaneously instead of one at a time, cutting training time from months to days or hours.
- Without this hardware, many modern AI breakthroughs (large language models, advanced image recognition) would be impractical to train.

4. Interview Questions

Q1. Why are GPUs preferred over CPUs for training deep learning models?

A: GPUs contain thousands of smaller cores designed for parallel processing, which matches the repetitive, parallel matrix calculations neural networks need — making training dramatically faster than a CPU's few, powerful sequential cores.

Cloud-Based AI

1. Introduction

Cloud-based AI means using remote, internet-hosted computing infrastructure — rather than your own local computer — to store data, train models, and run AI predictions. Major cloud providers offer ready-made AI infrastructure and tools on demand.



Amazon Web Services

Cloud AI, storage, and compute services



Google Cloud

AI/ML platform and data tools



Microsoft Azure

Enterprise cloud & AI services

2. What Is Cloud Computing (in AI context)?

Instead of buying and maintaining your own expensive GPU servers, you rent computing power and storage from a cloud provider over the internet — paying only for what you use, and scaling up or down instantly as your needs change.

3. Benefits of Cloud-Based AI

- **No large upfront cost** — avoid buying expensive GPU hardware.
- **Scalability** — instantly rent more computing power for large training jobs, then scale back down.
- **Access to pre-built AI tools** — ready-made APIs for vision, speech, and language tasks.
- **Global accessibility** — teams anywhere can access the same models and data securely.
- **Maintenance-free** — the provider handles hardware upkeep, security patches, and reliability.

4. Real-World Examples

Startups — Train and deploy AI models without owning a single physical server.

Streaming Services — Use cloud infrastructure to run recommendation models for millions of users simultaneously.

Mobile Apps — Call a cloud AI API to translate text or transcribe speech instantly, without heavy processing on the phone itself.

5. Interview Questions

Q1. Why do many companies prefer cloud-based AI over building their own AI infrastructure?

A: It avoids massive upfront hardware costs, offers instant scalability for training or serving models, and removes the burden of hardware maintenance — letting teams focus on building the AI product itself.

Summary, Comparison Table & Formulas

Summary of All Topics

- **Deep Learning** — Machine Learning using many-layered neural networks to learn patterns directly from raw data.
- **ML Training Process** — Data Collection → Preprocessing → Training → Evaluation → Tuning → Deployment.
- **Neural Network Architecture** — Input Layer receives data, Hidden Layers find patterns, Output Layer gives the answer.
- **Shallow vs Deep** — Shallow = few layers, simple problems; Deep = many layers, complex problems.
- **Applied vs Generalized AI** — Applied AI (real, task-specific) vs Generalized AI (theoretical, human-level).
- **Narrow vs Coordinated Tasks** — A single skill vs several narrow skills working together as a team.
- **Why AI Today** — Automation, big data handling, better decisions, improved accuracy.
- **Data in AI** — Structured (tables) vs Unstructured (text, images, audio) data.
- **Hardware in AI** — GPUs/TPUs enable fast, parallel training of neural networks.
- **Cloud-Based AI** — Renting scalable AI infrastructure from providers instead of owning hardware.

Master Comparison Table

Concept	Core Idea	Example
Shallow Learning	Few layers, simple patterns	Predicting exam score from study hours
Deep Learning	Many layers, complex patterns	Recognising objects in photos
Applied AI	One task, done very well	Voice assistant
Generalized AI	Any task, human-level (theoretical)	Not yet built
Structured Data	Organised rows/columns	Sales spreadsheet
Unstructured Data	Free-form content	Photos, audio, free text
GPU/TPU	Parallel processing hardware	Speeds up neural network training
Cloud AI	Rented, scalable infrastructure	AWS, Google Cloud, Microsoft Azure

Important Formulas & Concepts (Conceptual, Not Mathematical)

- **Neuron Output (conceptual):** $\text{Output} = \text{Activation}(\text{Input} \times \text{Weight}) + \text{Bias}$
- **Deep Learning Depth:** More Hidden Layers → More Abstraction → More Data/Compute Needed
- **Training Loop:** Predict → Compare to Correct Answer → Calculate Error → Adjust Weights → Repeat

Interview Questions (Beginner to Advanced) & MCQs

Interview Questions — Beginner to Advanced

1. (Beginner) What is Deep Learning?

A: A type of Machine Learning that uses neural networks with many layers to automatically learn patterns from raw data.

2. (Beginner) What are the three main layers in a neural network?

A: Input Layer, Hidden Layer(s), and Output Layer.

3. (Intermediate) What is the difference between a weight and a bias?

A: A weight controls how strongly one neuron's output influences the next neuron; a bias is an extra adjustable value added to help the neuron fit patterns more flexibly.

4. (Intermediate) Why is data preprocessing a critical step before training?

A: Raw data is often messy, incomplete, or inconsistent; cleaning and structuring it ensures the model learns accurate, meaningful patterns instead of noise.

5. (Intermediate) What's the difference between Applied AI and Generalized AI?

A: Applied AI is built for one specific task and is what exists today; Generalized AI would handle any intellectual task like a human, but remains theoretical.

6. (Advanced) Why do deep networks typically require GPUs instead of CPUs?

A: Training involves massive amounts of parallel matrix math; GPUs have thousands of small cores built for exactly this kind of parallel computation, making them far faster than CPUs for this workload.

7. (Advanced) What risk increases as you add more hidden layers to a network without adding more data?

A: Overfitting — the network may start memorising the training examples instead of learning patterns that generalise to new data.

8. (Advanced) Why might a company choose cloud-based AI infrastructure instead of on-premise hardware?

A: To avoid large upfront hardware costs, gain instant scalability for training/serving models, and offload hardware maintenance to the cloud provider.

15 MCQs with Answers

1. Deep Learning uses:

a) Fixed rules b) Multi-layered neural networks c) Spreadsheets only d) None — **Answer: b**

2. Which layer receives the raw input data?

a) Output Layer b) Hidden Layer c) Input Layer d) None — **Answer: c**

3. What does a "weight" control in a neural network?

a) Data storage b) Influence of one neuron on the next c) Screen brightness d) File size — **Answer: b**

4. Backpropagation is used to:

a) Collect data b) Adjust weights to reduce error c) Delete a model d) Store data in the cloud — **Answer: b**

5. Which comes first in the ML training process?

a) Deployment b) Data Collection c) Hyperparameter Tuning d) Evaluation — **Answer: b**

6. A shallow network typically has:

a) Many hidden layers b) 0-1 hidden layers c) No neurons d) Only output layers — **Answer: b**

7. Applied AI is also known as:

a) General AI b) Super AI c) Narrow AI d) None — **Answer: c**

8. Generalized AI (AGI) currently:

a) Exists in all smartphones b) Does not exist yet c) Is the same as a chatbot d) Was invented in 2010 — **Answer: b**

9. A self-driving car is an example of:

a) One single narrow skill b) Multiple coordinated narrow AI tasks c) Generalized AI d) Classical programming only — **Answer: b**

10. Which is an example of unstructured data?

a) A sales spreadsheet b) A customer ID table c) A voice recording d) A price list — **Answer: c**

11. GPUs are preferred for AI training because they:

a) Are cheaper than CPUs b) Perform many calculations in parallel c) Use less electricity d) Store more data — **Answer: b**

12. TPUs are specialised hardware built mainly for:

a) Web browsing b) Deep learning computations c) Word processing d) Gaming only — **Answer: b**

13. A key benefit of cloud-based AI is:

a) No internet needed b) Instant scalability without owning hardware c) It's always free d) It removes the need for data — **Answer: b**

14. Structured data is best described as:

a) Random text b) Organised rows and columns c) Audio waveforms d) Video frames — **Answer: b**

15. Why is AI needed today more than ever?

a) Data volumes and automation needs have grown massively b) Computers stopped working c) Humans no longer make decisions d) None — **Answer: a**

One-Page Quick Revision Sheet

Core Ideas

- **Deep Learning** = ML + many-layered neural networks learning patterns automatically.
- **Neural Network** = Input Layer → Hidden Layer(s) → Output Layer, connected by learnable weights.
- **Shallow** = few layers, simple tasks. **Deep** = many layers, complex tasks.
- **Applied AI** = real, task-specific (all of today's AI). **Generalized AI** = theoretical, human-level.
- **Narrow Task** = one skill. **Coordinated Tasks** = several narrow skills working as a team.

ML Training Pipeline

Data Collection → Preprocessing → Training → Evaluation → Hyperparameter Tuning → Deployment

Comparison At a Glance

Concept	Simple / Basic	Complex / Advanced
Network Depth	Shallow (few layers)	Deep (many layers)
AI Scope	Applied AI (one task)	Generalized AI (any task, theoretical)
Data Format	Structured (tables)	Unstructured (text/image/audio)
Compute Location	Local CPU	Cloud GPU/TPU clusters

Golden One-Liners

"Deep Learning stacks layers to learn deeper patterns. Applied AI does one job perfectly; Generalized AI (still theoretical) would do every job like a human. Structured data fits in tables; unstructured data needs deep learning to understand it. GPUs/TPUs and the cloud are what make training all of this practical at scale."